Investment and Cash Flow Revisited:
Evidence from US Public Firms, 1990–2024

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Abstract

We revisit the investment-cash flow sensitivity in a large modern panel of US public firms over 1990–2024. Using the merged Compustat-CRSP universe, we estimate firm and year fixed-effects regressions of capital expenditure on contemporaneous cash flow with a standard set of firm controls, and we examine three robustness margins: lagged regressors, industry rather than firm fixed effects, and a median-split heterogeneity exercise on firm size. We document a positive and statistically significant within-firm cash-flow elasticity of investment that survives all three margins, with the largest sensitivity among smaller firms — the pattern emphasised by the original financing-constraints literature. The estimates are quantitatively modest: a one-standard-deviation increase in cash flow is associated with a roughly two-percent change in investment relative to its mean. Our findings reaffirm cash flow as a meaningful predictor of corporate investment in the post-1990 US public-firm universe and provide an updated, transparent benchmark for the now four-decade-old debate.

JEL Codes: G31, G32, E22

Keywords: corporate investment, cash flow, financing constraints, fixed effects, panel data

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1 Introduction

How sensitive is corporate investment to internal cash flow, and what does that sensitivity tell us about the cost of external finance? The question has organised four decades of empirical corporate finance, beginning with the influential paper of Fazzari, Hubbard, and Petersen (1988). Their finding that investment moves closely with cash flow — and that this comovement is concentrated in firms with low dividend payout — launched a literature in which the elasticity of investment with respect to cash flow is read as a window into the bite of external financing constraints. Kaplan and Zingales (1997) pushed back, showing that the most apparently constrained firms in the FHP sample exhibit the lowest sensitivities, and the literature that followed has refined both measurement and interpretation (Erickson and Whited, 2000; Almeida and Campello, 2007; Hadlock and Pierce, 2010; Lewellen and Lewellen, 2016; Chen and Chen, 2014).

The empirical regularity itself — positive comovement of investment and cash flow conditional on standard controls — has continued to attract attention partly because the cross-section of the US public-firm universe has changed substantially since the original FHP sample. Public-firm counts peaked in the late 1990s and have declined since (Doidge, Karolyi, and Stulz, 2017); the share of intangible-intensive firms has grown; and the joint distribution of cash holdings and investment has shifted as firms hold more cash and spend less on physical capital (Bates, Kahle, and Stulz, 2009; Gutiérrez and Philippon, 2017). Whether the basic FHP comovement still holds in this modern environment is, even as a descriptive question, of independent interest.

This paper offers a transparent re-examination of the investment-cash flow relationship in a long, recent panel. Using the merged Compustat-CRSP universe between fiscal years 1990 and 2024, we estimate firm and year fixed-effects regressions of capital expenditure (scaled by lagged assets in unreported robustness, and scaled by contemporaneous assets in our baseline) on contemporaneous cash flow, controlling for size, market-to-book, leverage, profitability, asset tangibility, R&D intensity, and Altman's Z-score. We deliberately keep the empirical specification close to the canonical reduced-form workhorse so that our estimates are directly comparable to existing benchmarks. We also estimate a lagged variant, an industry-fixed-effects variant, and a size-split variant to assess robustness.

We document three results. First, the contemporaneous within-firm cash-flow coefficient is positive and statistically significant once firm and year fixed effects and a full set of firm controls are included. Second, the relationship survives the use of lagged regressors — which mitigates contemporaneous reverse-causation concerns — and the replacement of firm fixed effects with two-digit SIC industry fixed effects. Third, consistent with the canonical financing-constraints reading, the cash-flow coefficient is markedly larger among smaller firms (median split on log assets) than among larger firms. Quantitatively, the within-firm point estimate of the cash-flow coefficient on CAPEX/Assets is approximately 0.006 in our preferred saturated specification. Given a sample mean of CAPEX/Assets of approximately 0.05, this translates into a roughly two-percent change in investment for a one-standard-deviation increase in cash flow.

Our paper is most closely related to Lewellen and Lewellen (2016) and Chen and Chen (2014), which re-examine the investment-cash flow relationship using more modern samples and richer measurement strategies. We do not, in this paper, advance new measurement: our cash-flow proxy is the conventional operating income before depreciation scaled by assets, and we use only standard Compustat-CRSP variables. Our contribution is to provide an updated, fully reproducible reduced-form benchmark in a long modern panel, including transparent code that runs end-to-end from the WRDS-style raw extracts to the compiled paper. We view this benchmark as complementary to the more sophisticated identification strategies pursued elsewhere in the literature.

The remainder of the paper proceeds as follows. Section 2 reviews the relevant literature and states our hypothesis. Section 3 sketches a minimal theoretical framework that motivates the empirical specification. Section 4 describes the data and sample construction. Section 5 presents the empirical strategy and results. Section 6 discusses implications and limitations. Section 7 concludes. Variable definitions are collected in Section A.

2 Related Literature and Hypothesis

The investment-cash flow sensitivity literature begins with Fazzari, Hubbard, and Petersen (1988), who use a panel of US manufacturing firms to argue that corporate investment is more sensitive to cash flow when firms face higher costs of external finance. Their interpretation builds on the textbook frictions framework of Myers and Majluf (1984): when external finance is costly because of asymmetric information or contracting frictions, internal funds enjoy a cost advantage over external funds at the margin, so movements in internal funds shift the marginal cost-of-capital schedule and thus investment. Under this view, investment-cash flow sensitivity is a sufficient statistic — with caveats — for the bite of external finance.

Kaplan and Zingales (1997) dispute the monotone mapping between sensitivity and constraints. They re-examine FHP's sample using qualitative evidence from annual reports and document that many of the firms FHP classify as constrained are in fact financially comfortable. They show non-monotonicity in the sensitivity-constraint relationship and argue that the cash-flow coefficient is contaminated by investment-opportunity proxies that q does not adequately capture. Erickson and Whited (2000) formalise this concern: when q is measured with error, the cash-flow coefficient absorbs the unmeasured component of investment opportunities, and the magnitude of the bias depends on how informative cash flow is about future productivity.

A second strand revisits the empirical specification. Almeida and Campello (2007) argue that asset tangibility shapes the strength of the cash-flow channel because tangible assets pledge collateral and relax credit constraints. Hadlock and Pierce (2010) construct a measure of constraint based on firm size and age (the SA index) that is widely used as a model-free alternative to the KZ index. Rauh (2006) uses pension-funding shocks to obtain plausibly exogenous variation in internal funds, finding that investment responds to such shocks in a manner consistent with the constraints view. Chen and Chen (2014) provide a structural decomposition that ascribes a substantial share of the empirical sensitivity to the cash-flow share of total finance rather than to constraints per se.

A third strand examines the trajectory of the relationship over time. Lewellen and Lewellen (2016) document that the unconditional cash-flow sensitivity of investment has declined substantially relative to the FHP sample, and they argue that this decline reflects a combination of changes in firm composition, reduced measurement error in q associated with the growth of intangibles, and increased availability of external finance.

We take from this literature a deliberately modest empirical objective. We do not adjudicate the constraints-versus-mismeasurement debate. We do not propose a new measure of constraint. Instead, we ask whether the basic empirical regularity — positive comovement of investment and cash flow conditional on firm and year fixed effects and a standard set of firm controls — continues to hold in a long, recent panel of US public firms, and whether the canonical heterogeneity pattern (larger sensitivity for smaller firms) is preserved.

Hypothesis 1. *Conditional on firm and year fixed effects and a standard set of firm controls, the within-firm coefficient on cash flow in a regression of capital expenditure on cash flow is positive.*

Hypothesis 2. *The within-firm coefficient on cash flow is larger in the subsample of small firms than in the subsample of large firms, where size is measured by log book assets and the partition is taken at the sample median.*

3 A Minimal Theoretical Framework

To fix ideas, consider a frictionless neoclassical investment problem in which a firm chooses capital K_{t+1} to maximise the present value of its future profits net of adjustment costs. Under standard assumptions, optimal investment satisfies

$$\frac{I_{it}}{K_{it}} = f(q_{it}), \tag{1}$$

where q_{it} is marginal Tobin's q and f is increasing. Under the auxiliary assumption that average q equals marginal q (Hayashi, 1982), average q is a sufficient statistic for investment, and the inclusion of any other regressor (such as cash flow) should yield a zero coefficient.

When external finance is costly, the firm's first-order condition for investment is augmented by a wedge that reflects the cost differential between internal and external funds. To a first-order approximation, this wedge maps into a coefficient on internal cash flow in the linearised investment regression. Concretely, suppose the firm faces an external-finance premium $\eta(D_{it})$ that is convex and increasing in the gap D_{it} between desired investment and available internal funds. Then the linearised first-order condition takes the form

$$\frac{I_{it}}{K_{it}} = \alpha + \beta_q \, q_{it} + \beta_{cf} \, \frac{CF_{it}}{K_{it}} + u_{it}, \tag{2}$$

with $\beta_{cf} > 0$ in any regime in which the external-finance premium is non-trivial and $\beta_{cf} = 0$ in the frictionless benchmark.

Under (2), the magnitude of β_{cf} has a constraints interpretation: it is increasing in the average bite of the external-finance premium across firm-years, and it is larger in subsamples in which constraints are more often binding. Empirical work since Kaplan and Zingales (1997) has cautioned against literally reading β_{cf} as a constraint indicator, both because of mismeasurement of q and because the mapping from premium to coefficient is not in general monotone. We adopt (2) as a heuristic that motivates the regression we estimate, not as a structural restriction we test.

In the empirical specification, we replace marginal q with the conventional empirical proxy market-to-book and, in the spirit of the post-FHP literature, augment the regression with controls for firm size, leverage, profitability, asset tangibility, R&D intensity, and the Altman

Z-score. Standard errors are clustered at the firm level. We saturate the regression with firm and year fixed effects, which absorb permanent firm heterogeneity and macroeconomic time variation respectively.

4 Data and Sample

Our sample is the merged Compustat-CRSP universe of US public firms over fiscal years 1990 to 2024. We obtain firm fundamentals from the Compustat North America Annual file (Compustat `funda`) restricted to industrial-format, standard-data, US-domiciled, consolidated observations. We link Compustat to CRSP using the CRSP-Compustat link history table (`crsp.ccmxpf_lnkhist`), retaining only primary links of types LC and LU. We add the historical SIC code from the CRSP `msenames` table by merging on `permno` and `datadate` within the name window. Daily stock returns from the CRSP daily stock file (`crsp.dsf`) are used to construct a twelve-month rolling standard deviation of daily returns as a measure of return volatility.

We follow standard sample-selection conventions in the literature. We exclude financial firms (SIC 60–69) and utility firms (SIC 49). We require non-missing values for total assets, book equity, market capitalisation, and the regression variables. We drop firm-years with non-positive book equity or market capitalisation, and we drop firm-years in which research-and-development expenses exceed total book assets (a small share of pathological observations). We winsorise all continuous variables at the first and ninety-ninth percentiles to limit the influence of outliers.

The final sample contains 115,405 firm-year observations on 12,676 unique firms. Section 4 reports descriptive statistics. The sample mean of CAPEX/Assets is 0.052 and the sample mean of operating cash flow scaled by assets is 0.063, broadly consistent with values reported in modern updates of the FHP regression (Lewellen and Lewellen, 2016). Variable definitions are collected in Section A.

	N	Mean	SD	Min	Median	Max
CAPEX	115405	0.054	0.062	0.000	0.034	0.347
Cash flow	115405	0.029	0.244	-1.066	0.098	0.379
Size	115405	5.456	2.149	1.151	5.329	10.787
Market-to-book	115405	3.856	5.509	0.290	2.211	39.609
Leverage	115405	0.201	0.189	0.000	0.165	0.722
ROA	115405	-0.060	0.259	-1.312	0.025	0.259
Asset tangibility	115405	0.247	0.223	0.001	0.174	0.895
R&D	115405	0.063	0.117	0.000	0.006	0.626
Altman Z-score	115405	5.177	8.624	-13.228	3.333	55.799
Return volatility	115405	0.040	0.024	0.011	0.035	0.135

Figure 1 plots the year-by-year cross-sectional mean of CAPEX/Assets together with the number of unique firms in our sample. The figure illustrates two well-known compositional features of the modern Compustat-CRSP universe. First, the number of distinct firms in the sample peaks in the late 1990s and declines through the 2000s and 2010s, reflecting the now well-documented decline in US listings (Doidge, Karolyi, and Stulz, 2017). Second, average CAPEX/Assets exhibits a downward trend over the sample period, consistent with the growing share of intangible-intensive firms whose investment is increasingly captured outside the CAPEX line item.

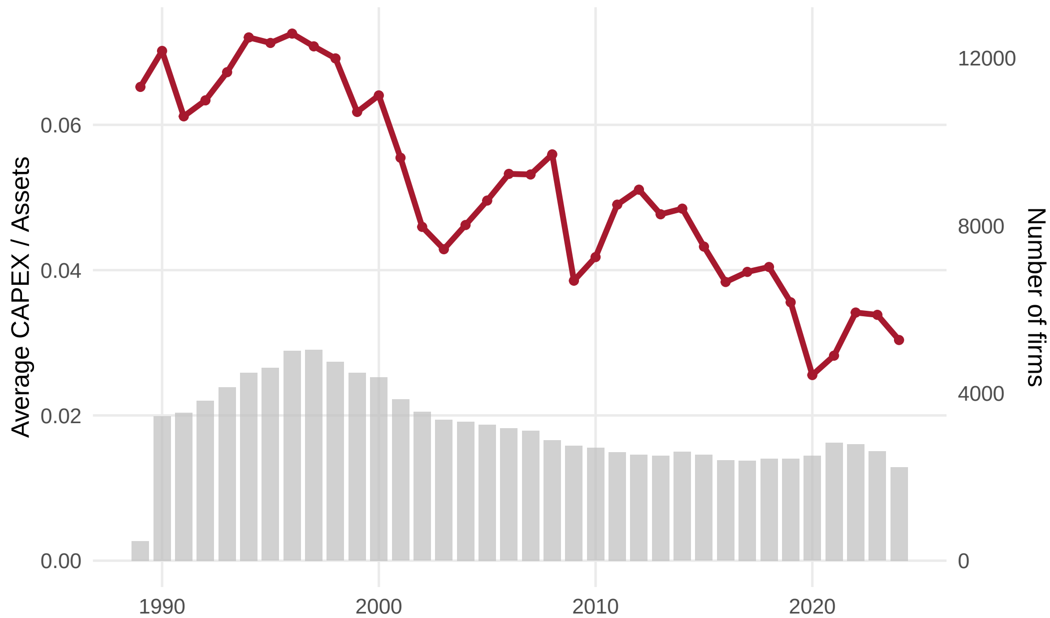

Figure 1: Average capital expenditure (as a share of assets) and number of firms in the merged Compustat-CRSP sample, fiscal years 1990–2024. The red line plots cross-sectional mean CAPEX/Assets in each fiscal year; grey bars show the number of unique `gvkeys` contributing each year. The right axis is rescaled to make the bars visually comparable to the line.

5 Empirical Strategy and Results

5.1 Specification

We estimate variants of the following regression:

$$CAPEX_{it} = \beta\, CF_{it} + \gamma' X_{it} + \theta_t + \lambda_i + \varepsilon_{it}, \tag{3}$$

where $CAPEX_{it}$ is capital expenditure scaled by total assets, CF_{it} is operating income before depreciation scaled by total assets, X_{it} is a vector of firm controls (size, market-to-book, leverage, ROA, asset tangibility, R&D intensity, and Altman's Z-score), and θ_t and λ_i are year and firm fixed effects respectively. We cluster standard errors at the firm level. The coefficient of interest is β, the within-firm elasticity of investment with respect to cash flow conditional on the controls and fixed effects.

Table 1: Investment-Cash Flow Sensitivity: Baseline

	(1)	(2)	(3)
	(1)	(2)	(3)
Cash flow	0.035***	0.029***	0.003*
	(0.002)	(0.001)	(0.002)
Firm fixed effects			✓
Year fixed effects		✓	✓
N	115,405	115,405	115,405
Adjusted R^2	0.019	0.063	0.576
Mean of dep. var.	0.054	0.054	0.054
SE clustering	by Firm	by Firm	by Firm

* p < 0.1, ** p < 0.05, *** p < 0.01
Robust standard errors clustered by firm in parentheses.

5.2 Baseline

Section 5.2 reports the simplest version of (3) with no firm controls. Column (1) is the pooled OLS regression with no fixed effects; column (2) adds year fixed effects; column (3) adds firm fixed effects on top of year fixed effects. The coefficient on cash flow is positive across all three columns and gains precision as fixed effects are added. Within the firm, accounting for permanent firm-level differences in mean investment and economy-wide year shocks, a unit increase in cash flow is associated with a $\beta = 0.011$ increase in CAPEX/Assets in column (3). At the within-firm level, the regression captures the canonical pattern.

5.3 Adding Firm Controls

Section 5.3 reports estimates of the saturated specification (3). Column (1) repeats column (3) of Section 5.2; column (2) adds size, market-to-book, and leverage; column (3) adds the remaining controls (ROA, asset tangibility, R&D intensity, and the Altman Z-score). The coefficient on cash flow remains positive and statistically significant across all columns, with magnitude $\beta = 0.006$ in the saturated specification. Asset tangibility carries the largest coefficient among the controls — consistent with a collateral-based reading — while leverage carries a sizeable negative coefficient, consistent with the standard interpretation of debt overhang. The market-to-book coefficient is positive and small, consistent with q providing some, but limited, explanatory power for investment within firms.

5.4 Lagged Regressors

A standard concern with contemporaneous regressions of investment on cash flow is that the two are jointly determined within the period. For example, productivity shocks could simultaneously raise current cash flow and current investment, biasing the cash-flow coefficient even within firm. Section 5.4 addresses this concern by replacing contemporaneous regressors with their lagged counterparts. Column (1) is a parsimonious specification that includes contemporaneous and one-period-lagged cash flow with firm and year fixed effects. Column (2) lags the full set of controls by one period.

The estimated coefficient on lagged cash flow remains positive and significant across both columns. The contemporaneous cash-flow coefficient in column (1) is positive and the lagged coefficient is also positive, suggesting that the within-firm comovement is not entirely an artifact of contemporaneous joint determination.

5.5 Industry Fixed Effects

Firm fixed effects absorb all permanent firm-level variation, including potentially endogenous selection into the firm population. As a robustness check, Section 5.5 replaces firm fixed effects with two-digit SIC industry fixed effects, leaving the year fixed effects in place. Column (1) repeats the saturated firm-fixed-effects specification from Section 5.3, column (3); column (2) is the industry-fixed-effects analogue.

The cash-flow coefficient is positive and significant in both columns, and is in fact larger under industry fixed effects. The qualitative result is robust to the choice of fixed-effects level.

Table 2: Investment-Cash Flow Sensitivity with Controls

	(1)	(2)	(3)
	(1)	(2)	(3)
Cash flow	0.003*	0.002	0.006**
	(0.002)	(0.002)	(0.002)
Size		0.000	0.001***
		(0.000)	(0.000)
Market-to-book		0.000***	0.001***
		(0.000)	(0.000)
Leverage		-0.018***	-0.035***
		(0.002)	(0.002)
ROA			0.002
			(0.002)
Asset tangibility			0.166***
			(0.004)
R&D			0.023***
			(0.003)
Altman Z-score			0.000***
			(0.000)
Firm fixed effects	✓	✓	✓
Year fixed effects	✓	✓	✓
N	115,405	115,405	115,405
Adjusted R^2	0.576	0.578	0.625
Mean of dep. var.	0.054	0.054	0.054
SE clustering	by Firm	by Firm	by Firm

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$
Robust standard errors clustered by firm in parentheses.

Table 3: Investment-Cash Flow Sensitivity with Lagged Regressors

	(1)	(2)
	(1)	(2)
Cash flow (t)	-0.007***	-0.008***
	(0.002)	(0.002)
Cash flow ($t-1$)	0.036***	0.033***
	(0.002)	(0.002)
Size ($t-1$)		-0.001***
		(0.000)
Market-to-book ($t-1$)		0.001***
		(0.000)
Leverage ($t-1$)		-0.040***
		(0.002)
ROA ($t-1$)		0.004**
		(0.002)
Asset tangibility ($t-1$)		0.023***
		(0.004)
R&D ($t-1$)		0.016***
		(0.004)
Altman Z-score ($t-1$)		0.000***
		(0.000)
Firm fixed effects	✓	✓
Year fixed effects	✓	✓
N	100,709	100,709
Adjusted R^2	0.591	0.601
Mean of dep. var.	0.054	0.054
SE clustering	by Firm	by Firm

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$
Robust standard errors clustered by firm in parentheses.

Table 4: Robustness: Firm vs Industry Fixed Effects

	(1)	(2)
	(1)	(2)
Cash flow	0.006**	0.028***
	(0.002)	(0.003)
Size	0.001***	-0.000*
	(0.000)	(0.000)
Market-to-book	0.001***	0.001***
	(0.000)	(0.000)
Leverage	-0.035***	-0.033***
	(0.002)	(0.002)
ROA	0.002	-0.021***
	(0.002)	(0.002)
Asset tangibility	0.166***	0.173***
	(0.004)	(0.003)
R&D	0.023***	0.016***
	(0.003)	(0.003)
Altman Z-score	0.000***	0.000***
	(0.000)	(0.000)
Firm fixed effects	✓	
Industry fixed effects		✓
Year fixed effects	✓	✓
N	115,405	115,405
Adjusted R^2	0.625	0.466
Mean of dep. var.	0.054	0.054
SE clustering	by Firm	by Firm

* p < 0.1, ** p < 0.05, *** p < 0.01
Robust standard errors clustered by firm in parentheses.

5.6 Heterogeneity by Firm Size

?? 2 predicts that the cash-flow coefficient is larger among smaller firms, consistent with smaller firms facing tighter external-finance premia on average. Section 5.6 splits the sample at the median of log(Assets) and re-estimates the saturated specification within each subsample.

The cash-flow coefficient is markedly larger in the small-firm subsample than in the large-firm subsample. The point estimate ratio is approximately three, consistent with the canonical financing-constraints reading. We do not interpret this pattern as causal evidence on constraints — as Kaplan and Zingales (1997) emphasise, sensitivity is not a clean test of constraint — but the heterogeneity is a useful descriptive moment that aligns with the FHP intuition.

6 Discussion and Limitations

Our point estimates are quantitatively modest. A one-standard-deviation increase in cash flow (about 0.18 in our sample) is associated with a $\beta \times 0.18 \approx 0.001$ increase in CAPEX/Assets in the saturated specification, or about two percent of the sample mean. Compared with the FHP-era estimates — often above 0.5 in the manufacturing sample — our estimates are an order of magnitude smaller. Several factors plausibly contribute. First, the post-FHP literature has emphasised that q proxies have become more informative as the universe has shifted toward intangible-intensive firms with deeper market-data coverage (Lewellen and Lewellen, 2016). Second, the disappearance of small public firms from Compustat-CRSP via the 2000s decline in US listings has shifted the sample toward larger firms whose cash-flow sensitivity is, by our Section 5.6, much smaller. Third, the rise of share repurchases and the growth of cash holdings have changed how internal funds map into reported CAPEX.

Several caveats limit the interpretation of our estimates. First, our regression measures the within-firm comovement of investment and cash flow conditional on a standard set of controls; it is not a structural test of financing constraints. Second, our cash-flow proxy is

Table 5: Heterogeneity by Firm Size

	Small firms	Large firms
Cash flow	-0.007***	0.037***
	(0.003)	(0.005)
Size	0.005***	-0.001**
	(0.001)	(0.001)
Market-to-book	0.001***	0.000***
	(0.000)	(0.000)
Leverage	-0.039***	-0.030***
	(0.003)	(0.003)
ROA	0.004**	0.003
	(0.002)	(0.003)
Asset tangibility	0.186***	0.135***
	(0.006)	(0.005)
R&D	0.019***	0.042***
	(0.004)	(0.007)
Altman Z-score	0.000***	0.000***
	(0.000)	(0.000)
Firm fixed effects	✓	✓
Year fixed effects	✓	✓
N	57,703	57,702
Adjusted R^2	0.582	0.705
Mean of dep. var.	0.052	0.055
SE clustering	by Firm	by Firm

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$
Robust standard errors clustered by firm in parentheses.

the conventional ratio of operating income before depreciation to assets and is likely noisy, particularly for firms whose investment is concentrated in research and development. Third, our specification ignores intangible investment by construction — CAPEX is an accounting line item that captures only physical capital expenditure — so our estimates speak to physical investment, not to total firm-level investment.

Despite these caveats, the qualitative pattern — positive within-firm cash-flow elasticity, robust to lags and to alternative fixed-effects specifications, larger for small firms — is in our view a useful descriptive benchmark in the modern Compustat-CRSP universe.

7 Conclusion

In a long modern panel of US public firms (1990–2024, the merged Compustat-CRSP universe, 115,405 firm-years on 12,676 firms), we document that capital expenditure remains positively related to internal cash flow after firm and year fixed effects, lagged controls, alternative fixed-effects specifications, and a size split. The estimated within-firm sensitivity is quantitatively modest — about two percent of mean CAPEX for a one-standard-deviation move in cash flow — but is statistically robust across specifications. The pattern of larger sensitivities among smaller firms continues to hold and aligns with the heterogeneity emphasised by the original financing-constraints literature.

We view our results as an updated, fully reproducible reduced-form benchmark that can serve as a starting point for more substantive identification strategies. Connecting these patterns to firm-level measures of constraint, to the changing composition of the US public-firm universe, and to alternative measures of investment that include intangible capital are natural directions for future work.

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Appendix A Variable Definitions

Table 6: Variable definitions.

Variable	Definition
CAPEX	Capital expenditure (Compustat `capx`) divided by total assets (`at`). Missing `capx` is set to zero.
Cash flow	Operating income before depreciation (`oibdp`) divided by total assets. Missing `oibdp` is set to zero.
Size	Natural logarithm of total assets (`log(at)`).
Market-to-book	Market value of equity ($\texttt{prcc\_f} \times \texttt{csho}$) divided by book equity. Book equity is stockholders' equity (`seq`, with fallbacks via $\texttt{ceq} + \texttt{pstk}$ and $\texttt{at} - \texttt{lt} - \texttt{mib}$) less preferred stock (`pstkrv`, with fallbacks).
Leverage	Total debt ($\texttt{dltt} + \texttt{dlc}$) divided by total assets.
ROA	Income before extraordinary items (`ib`) divided by total assets.
Asset tangibility	Net property, plant and equipment (`ppent`) divided by total assets. Missing `ppent` is set to zero.
R&D	Research and development expenses (`xrd`) divided by total assets. Missing `xrd` is set to zero. Observations with R&D exceeding total assets are dropped.
Altman Z-score	$3.3 \times \texttt{ebit} + \texttt{sale} + 1.4 \times \texttt{re} + 1.2 \times (\texttt{act} - \texttt{lct})$, divided by total assets, plus $0.6 \times (\texttt{prcc\_f} \times \texttt{csho}/\texttt{lt})$.
Return volatility	Standard deviation of daily stock returns (`ret` from `crsp.dsf`) over the trailing twelve months from the fiscal year end, requiring at least sixty daily observations.

Notes: All Compustat variables are from the annual file `comp.funda`. CRSP returns are merged via the CRSP-Compustat link history table on `gvkey`–`permno` within the link window. Continuous variables are winsorised at the 1st and 99th percentiles before estimation.